

Navigating the Filesystem Using a Terminal

- \$ **cd** ~
- The ~ character can change current directory to home directory.
- \$ **pwd**
- o/p: /home/pi
- confirm that the command did indeed set the directory to the home directory using the pwd command

- `$ cd ..`
- move up one level in the directory structure
- `$ cd /`
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- Check out the root of the directory structure
- `$ nano myfile.txt`
- Create file using nano editor and edit file

- `$ cp myfile.txt myfile2.txt`
- copy of a file within working directory.
- `$ echo "hello" > myfile.txt`
- `$ ls`
- `myfile.txt`
- `$ cp myfile.txt myfile2.txt`
- `$ ls`
- `myfile.txt myfile2.txt`
- Use the `>` and `echo` commands to redirect what you type on the command line to a file.
- to create a one-line file without having to use an editor.

- `$ mv my_file.txt my_file.rtf`
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- simply rename a file from `my_file.txt` to `my_file.rtf`
- `$ more myfile.txt`
- You want to view the contents of a small file without editing it.
- The `more` command just displays one screen of text at a time

- `$ mkdir my_directory`
- To create a directory, use the `mkdir` command.
- `$ rm my_file.txt`
- `$ rm my_directory`
- Deleting a single file is simple and safe. The following example will delete the file `my_file.txt` from the home directory
- `$ rm -r mydir`
- to recursively delete a directory and all its contents, including any directories that it contains.
- `$ rm *`
- delete all the files in the directory.

Performing Tasks with Superuser Privileges

- Some commands don't work because user have insufficient privileges. user need to issue commands with superuser privileges.
- The sudo (superuser do) command allows user to perform actions with superuser privileges. Just prefix the command with **sudo**.

- The apt-get package manager uses a list of available software. This list is included with the Raspberry Pi operating system distribution that you use, but is likely to be out-of-date. So, if the software that you try to install is reported by apt-get as not found, run
 - `$ sudo apt-get update`
 - `$ sudo apt-get upgrade`

- `$ sudo apt-get install <name of software>`
 - `$ sudo apt-get install abiword`
 - The most used tool for installing software from a Terminal session is apt-get.
 - The basic format of the command, which you must run as a superuser.
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- `$ sudo apt-get remove <name of software>`
 - `$ sudo apt-get remove abiword`
 - The apt-get utility has an option (remove) that will remove a package, but only packages that have been installed with apt-get install.

- Removing a package is not always delete everything, as packages often also have prerequisite packages that are installed as well. To remove these, user can use the **autoremove** option, as shown here:
- **\$ sudo apt-get autoremove <software name>**
- **\$ sudo apt-get autoremove abiword**
- **\$ sudo apt-get clean**
- The apt-get clean option will do some further tidying up of unused package installation files.

- The wget command use to download a file from the Internet without using a web browser.
- `$ wget <URL>`
- `$ wget`
<http://www.icrobotics.co.uk/wiki/images/c/c3/Pifm.tar.gz>
- `$ sudo apt-get install git`
- `$ git clone`
https://github.com/simonmonk/raspberrypi_cookbook.git
- To use code in Git repositories, you need to download Git and then use the git clone command to fetch the files.

- **\$ history**

- command history will grow too large to find a command that you used ages ago. To find a command from way back, you can use the history command.
- You can access the previous command you ran by pressing the up key. Pressing it again will take you to the command before that, and so on. If you overshoot the command you wanted, the down arrow key will take you back in the other direction

- `$ gunzip myfile.gz`
- `$ tar -xzf myfile.tar.gz`
- the tar or gunzip commands use to uncompress the downloaded compressed files.
- `$lsusb`
- the lsusb command. This will list all the devices attached to the USB ports on Raspberry Pi:

- `$ cat file1.txt file2.txt file3.txt > full_file.txt`
- the cat command to concatenate a number of files into one output file.
- `$ date`
- date command use to display the UTC time.
- The date and time format is `MMDDHHMMYYYY`, where MM is the month number, DD the day of the month, HH and MM the hours and minutes, respectively, and YYYY the year.

Assignment

- chmod
- kill
- chown
- df
- alias
- pipes
- crontab